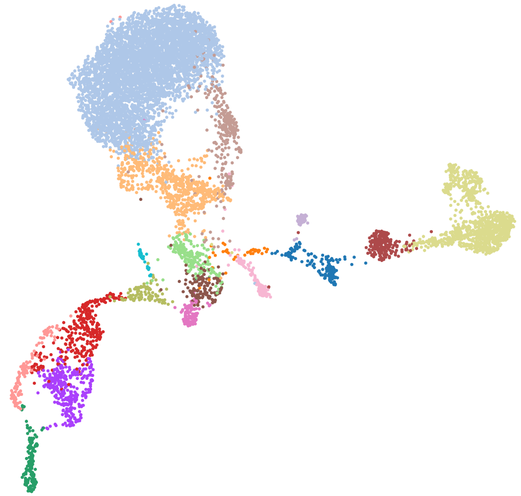
ViewmastR: de novo celltyping and refinement based on transfer learning from reference annotations

(a) Reference: BoneMarrowMap (118,509 real cells)



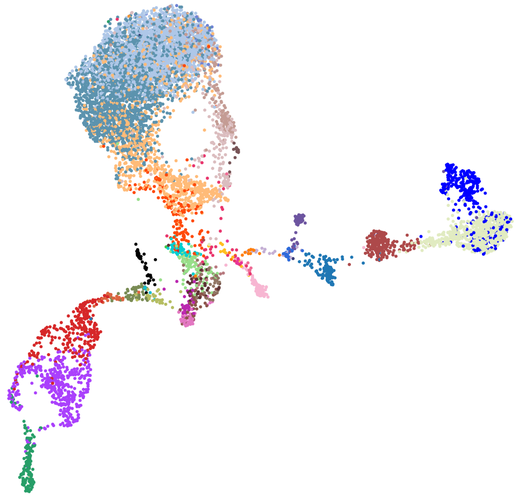
HSC
MPP-MyLy
MPP-MkEry
LMPP
Cycling Progenitor
MLP
MLP-II
CLP
Pre-ProB
Pro-B VDJ
Pro-B Cycling
Large Pre-B
Small Pre-B
Immature B
Mature B
Plasma Cell
Early GMP
GMP-Cycle
GMP-Neut
GMP-Mono
Early ProMono
Late ProMono
CD14 Mono
CD16 Mono
Pre-cDC
cDC1
cDC2
ASDC
Pre-pDC
Pre-pDC Cycling
pDC
EoBasoMast Precursor
MEP
Megakaryocyte Precursor
Megakaryocyte
BFU-E
CFU-E
Pro-Erythroblast
Basophilic Erythroblast
Polychromatic Erythroblast
Orthochromatic Erythroblast
CD4 Naive
CD4 Central Memory
CD4 Effector Memory
CD4 Regulatory
CD8 Naive
CD8 Central Memory
CD8 Effector Memory 1
CD8 Effector Memory 2
CD8 Tissue Resident Memory
T Proliferating
NK CD56high
NK
NK Proliferating
Stromal

(b) Query: authors' labels (8,627 cells)



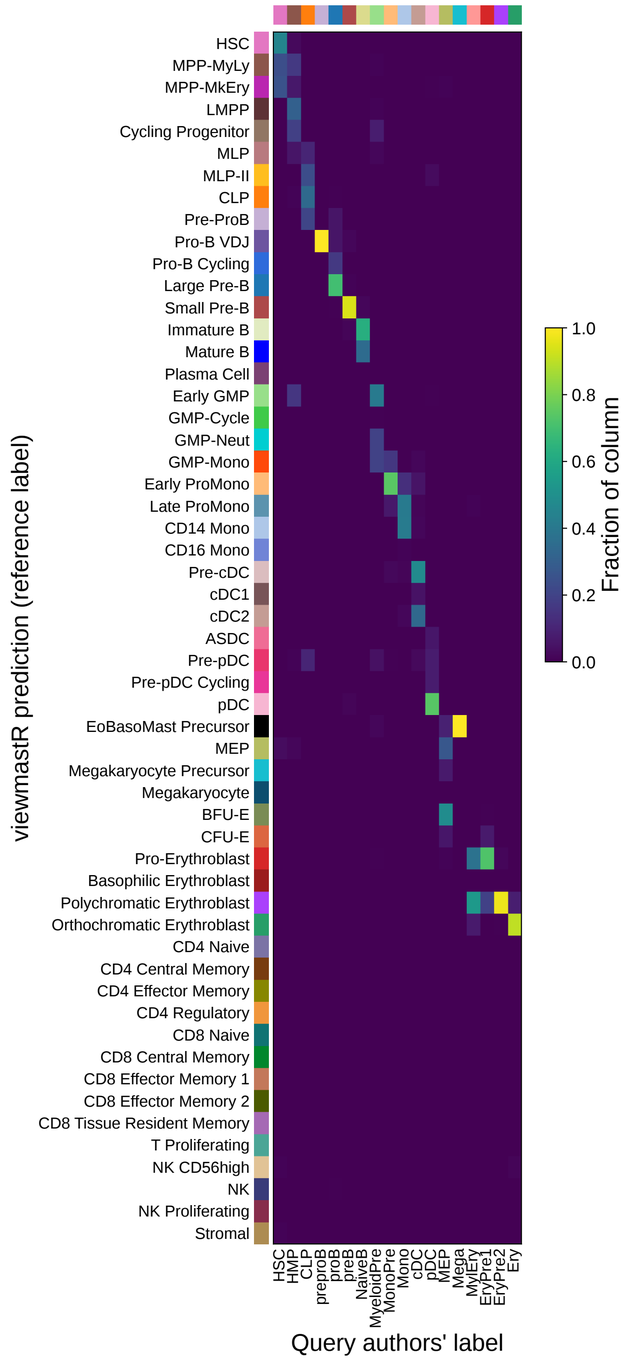
HSC
HMP
CLP
preproB
proB
preB
NaiveB
MyeloidPre
MonoPre
Mono
cDC
pDC
MEP
Mega
MyEry
EryPre1
EryPre2
Ery

(c) Query: viewmastR prediction (reference labels)

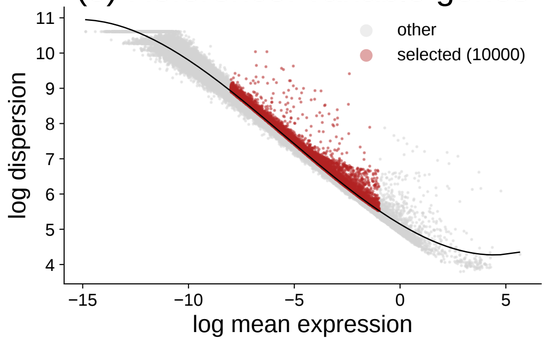


HSC
MPP-MyLy
MPP-MkEry
LMPP
Cycling Progenitor
MLP
MLP-II
CLP
Pre-ProB
Pro-B VDJ
Pro-B Cycling
Large Pre-B
Small Pre-B
Immature B
Mature B
Early GMP
GMP-Neut
GMP-Mono
Early ProMono
Late ProMono
CD14 Mono
Pre-cDC
cDC1
cDC2
Pre-pDC
Pre-pDC Cycling
pDC
EoBasoMast Precursor
MEP
BFU-E
CFU-E
Pro-Erythroblast
Polychromatic Erythroblast
Orthochromatic Erythroblast

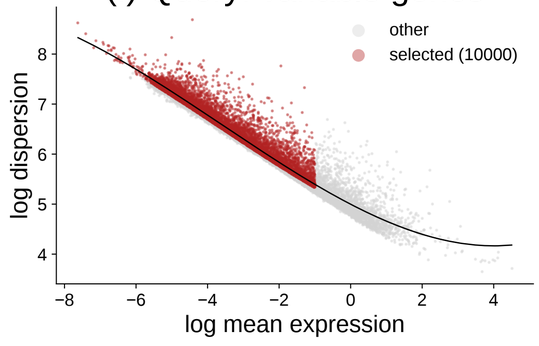
(d) Prediction vs authors' label



(e) Reference: variable genes



(f) Query: variable genes



3,878 common genes used for transfer